# Chapter 4 R Practice

## Uniform distribution: Suppose $X \sim U(0, 5)$.

1. Sketch the pdf and shade the region $X < 2$

2. Write out the integral and the R code required to calculate $P(X < 2)$. Calculate it using your preferred method.

3. Write out two different lines of code you could use to find $P(X > 2)$ in R. *Hint: one should make use of the `lower.tail` argument.* Execute both lines of code in R to verify that they give the value you expect.

4. Suppose we want to find $P(2 < X < 3)$. Sketch a graph with this region shaded.

5. Using your visual as a guide, what would you need to subtract from $P(X < 3)$ in order to obtain $P(2 < X < 3)$?

6. Using your answer to part 5 as a guide, write out R code for computing $P(2 < X < 3)$. Execute your code in R and report the answer.

## Normal distribution

7. Recall that Z-scores are normally distributed with mean 0 and standard deviation 1. Find each of the following probabilities using R code. Note, you should draw a picture to help you visualize what you are trying to find for each question

   a) P(Z $<$ 1.25)

   b) P(Z $>$ 1.25)

   c) P(Z $<$ -1.25)

   d) P(Z $>$ -1.25)

   e) P(-1 $<$ Z $<$ 1)

   f) P(-2 $<$ Z $<$ 2)

   g) P(-3 $<$ Z $<$ 3)

   h) P(-1.645 $<$ Z $<$ 1.645)

   i) P(-1.96 $<$ Z $<$ 1.96)

   j) P(-2.576 $<$ Z $<$ 2.576)

8. Use the qnorm() function to find:

   a) the 95th percentile of Z-scores

   b) the 5th percentile of Z-scores

   c) the 2.5th percentile of Z-scores

   d) the 97.5th percentile of Z-scores

   e) the 99.5th percentile of Z-score

9. Suppose that birthweights of newborn babies in the United States follow a normal distribution with mean 3300 grams and standard deviation 500 grams. Babies who weigh less than 2500 grams at birth are classified as low birthweight. Use R code to answer each of the following:

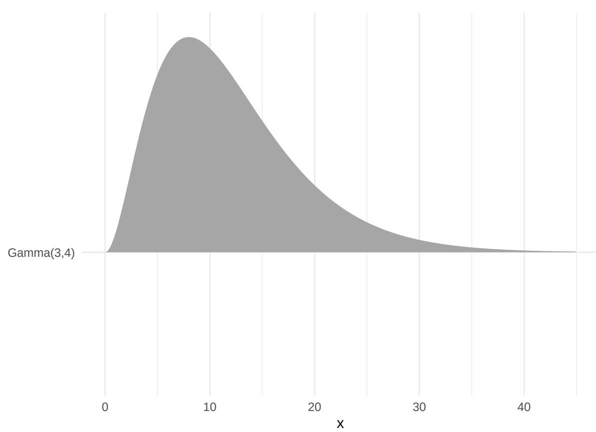
   a) What percentage of newborn babies weigh less than 2500 grams?
   b) What is the probability that a randomly selected newborn baby weighs more than 10 pounds?
   c) What percentage of newborn babies weigh between 3000 and 4000 grams?
   d) How little must a baby weight to be among the lightest 2.5% of all newborns? Convert your answer to pounds. Google to figure out the conversion!
   e) How much must a baby weigh to be among the heaviest 10%? Convert your answer to pounds.

#### Gamma example

Suppose $X \sim \Gamma(\alpha = 3, \beta = 4)$.

We can plot this distribution with the following code:

```
ggplot() +
  stat_dist_slab(aes(y = "Gamma(3,4)",
                              #note the function dist_gamma() requires the rate parameter
                              #instead of the scale parameter. Recall rate = 1/scale
                              dist = dist_gamma(shape = 3, rate = 1/4))) +
  labs(y = "") +
  theme_minimal()
```




10. What's the mean of this distribution? Does this seem reasonable based on the graph above?

11. Shade the region $X < 5$ on the graph above

12. Write out the code you need to calculate $P(X < 5)$. *Hint: use the `pgamma()` function.*

13. Use your code above to compute $P(X < 5)$ and report the answer. Does the value seem reasonable based on the plot above?

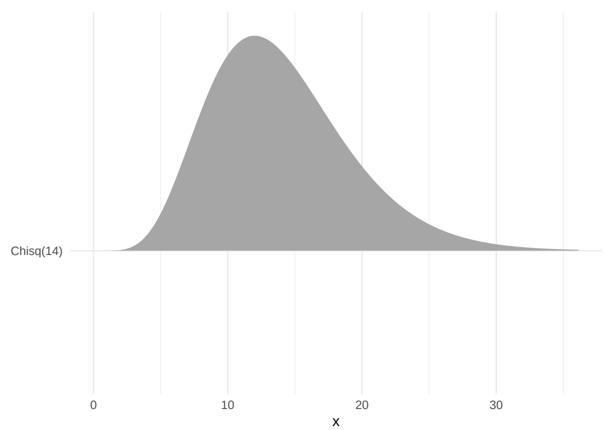
14. Write out two different lines of code you could use to find $P(X > 5)$ in R. *Hint: one should make use of the `lower.tail` argument.* Execute both lines of code in R to verify they give the value you expect.

### Chi-squared example

Suppose $X \sim \chi^2(14)$.

We can plot this distribution with the following code:

```
ggplot() +
  stat_dist_slab(aes(y = "Chisq(14)",
                     dist = dist_chisq(14))) +
  labs(y = "", y = "y") +
  theme_minimal()
```




15. What's the mean of this distribution? Does this seem reasonable based on the graph above?

16. Shade the region $X < 10$ on the graph above

17. Write out the code you need to calculate $P(X < 10)$. *Hint: use the* `pchisq()` *function.*

18. Use your code above to compute $P(X < 10)$ and report the answer. Does the value seem reasonable based on the plot above?

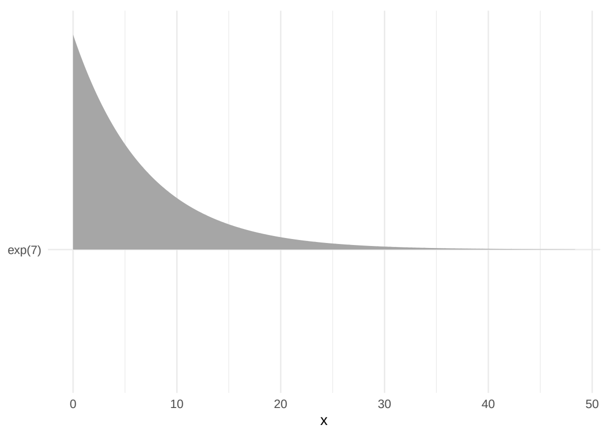
19. Write out two different lines of code you could use to find $P(X > 10)$ in R. *Hint: one should make use of the* `lower.tail` *argument.* Execute both lines of code in R to verify they give the value you expect.

**Exponential example**

Suppose $X \sim exp(7)$.

We can plot this distribution with the following code:

```
ggplot() +
  stat_dist_slab(aes(y = "exp(7)",
                             #note the function dist_exponential() requires the rate parameter
                             #instead of the scale parameter. Recall rate = 1/scale
                             dist = dist_exponential(1/7))) +
  labs(y = "", y = "y") +
  theme_minimal()
```




20. What's the mean of this distribution? Does this seem reasonable based on the graph above?

21. Write out the code you need to calculate $P(X > 20)$. *Hint: use the* `pexp()` *function.* How many/what arguments do you need to specify?

22. Use your code above to compute $P(X > 20)$ and report the answer. Does the value seem reasonable based on the plot above?

23. Shade $P(10 < X < 20)$ on the plot above.

24. Write out the code you need to find $P(10 < X < 20)$. Execute the code and report the answer. Does it seem reasonable based on the plot above?

## Computing percentiles for continuous distributions in R

The inverse of the cdf, or the $x$ that solves the equation $p = F(x) = P(X < x)$, are returned with commands using "q" at the beginning of the R distribution name:

| Distribution | Quantile Function | Connection to textbook parametrization |
|---|---|---|
| Normal | `qnorm(p, mean, sd)` | $\mu, \sigma$ |
| Uniform | `qunif(p, min, max)` | $\theta_1, \theta_2$ |
| Exponential | `qexp(p, rate)` | `rate` $= \lambda = 1/\beta$ |
| Gamma | `qgamma(p, shape, rate)` | `shape` $= \alpha$, `rate` $= \lambda = 1/\beta$ |
| Chi-square | `qchisq(p, df)` | $\nu$ |
| Beta | `qbeta(p, shape1, shape2)` | $\alpha, \beta$ |

Again, set `lower.tail = FALSE` to use upper tail probabilities and find $x$ such that $P(X > x) = p$.

### Gamma example

Suppose $X \sim \Gamma(\alpha = 3, \beta = 7)$. Below is a visual of the distribution shaded at the median.


25. Write out the code you need to find the median in R. Execute the code and report the value. Does the value seem reasonable based on the plot above?

26. Write out the code you need to find the 95th percentile in R. Execute the code and report the value. Does the value seem reasonable based on the plot above?